

Health Information Technology Supporting Adherence Memory Disorder Patients: A Systematic Literature Review

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Abstract

Background People with memory disorders have difficulty adhering to treatments. With technological advances, it remains important to investigate the potential of health information technology (HIT) in supporting medication adherence among them.

Objectives This review investigates the role of HIT in supporting adherence to medication and therapies among patients with memory issues. It also captures the factors that impact technology adherence interventions.

Methods We searched the literature for relevant publications published until March 15, 2023, using technology to support adherence among patients with memory issues (dementia, Alzheimer's, amnesia, mild cognitive impairment, memory loss, etc.). The review was reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines. We conducted a quality assessment of the papers following the Mixed Methods Appraisal Tool.

Results Fifteen studies were included after carefully reviewing the 3,773 articles in the search. Methodological quality, as appraised, ranged from 80 to 100% with eight studies rated 100%. The studies overall did not have a high risk of bias. Thus, all of the 15 studies were included. Technologies investigated were classified into four groups based on their impact: therapeutic patient education, simplifying treatment regimens, early follow-up visits and short-term treatment goals, and reminder programs. Different technologies were used (automatic drug dispensers or boxes, mobile health-based interventions, game-based interventions, e-health-based interventions, patient portals, and virtual reality). The factors impacting patients' adherence to technology-based treatment and medication were clustered into human–computer interaction and integration challenges.

Conclusion This study contributes to the literature by classifying the technologies that supported medication adherence among patients with memory issues in four groups. It also explores and presents the possible limitations of existing solutions to drive future research in supporting care for people with memory disorders.

Keywords

- ▶ health information technology
- ▶ memory disorder
- ▶ Alzheimer's
- ▶ dementia
- ▶ medication adherence

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Background and Significance

Memory disorder issues are unusual forgetfulness that prevents people from remembering new events, recalling memories of the past, or both.^{1,2} This decline in memory can be associated with age, or, in severe cases, with diseases such as Alzheimer's, dementia, or other brain disorders.³ This problem is very common globally. For instance, the Alzheimer's Association International Conference reported in 2021 that each year, 10 in every 100,000 individuals develop dementia with early onset.⁴ This is the equivalent of 350,000 new-onset cases every year globally.⁴ Additionally, the global rate of dementia is estimated to increase from 57.4 million cases in 2019 to 152.8 million cases in 2050.⁵ In the United States specifically, the mortality rate from Alzheimer's increased in the overall population from 16 per 100,000 in 1999 to 30 in 2019, equivalent to an 88% increase in rate.^{4,5}

Memory loss has serious consequences on patients' lives. Because of their difficulties, people can begin withdrawing from social relationships and activities.⁶ They may also experience various psychological symptoms (e.g., agitation, depression) that can negatively impact their behavior and quality of life.⁷ But mainly, impairment of this cognitive function may substantially compromise the person's medication adherence behavior, resulting in discontinuous access to care (problems with drug adherence, medical engagement, persistence, etc.).^{8,9} Medication adherence is the extent to which a patient's behavior corresponds to the health care goals established by his doctors, including taking medication and executing a lifestyle change.^{9–12} It can be evaluated based on the patient's respect for doses and how much medication was prescribed (compliance) or for the therapy continuity and duration over time (persistence).¹⁰

Both compliance and persistence affect directly the clinical outcomes.¹⁰ Medication adherence can affect quality and length of life, health outcomes, and overall health care costs.^{13,14} Nonadherence can account for up to 50% of treatment failures, around 125,000 deaths, and up to 25% of hospitalizations each year in the United States.^{13,14} Considering the high risk of nonadherence among patients with memory decline, it remains important to support their adherence to treatment. A review by Smith et al highlighted the importance of medication adherence among people with dementia or cognitive impairment suggesting the need for more diving into the possible ways to support compliance and persistence among this population.¹⁵

Over the last decades, the approaches aimed to improve the continuity of therapies among patients with chronic diseases have witnessed major developments focusing on patients, providers' roles, systems' design, and environments' management.⁹ With the widespread use of technology in health care, health information technologies have become promising medication adherence aids thanks to simplicity, user-friendliness, and accessibility for the public.¹⁶ For instance, different mHealth applications were shown to be effective in supporting medication adherence.^{16,17} A study by Peng et al showed that, compared with conventional care, mHealth interventions, such as short message service and electronic pillboxes, are

shown to have better effects on medication adherence.¹⁸ Many reviews have also highlighted the potential that technology has in supporting medication adherence.^{19–22} Yet, most of them focus on general populations or on patients with chronic diseases.^{17,23} To our knowledge, no review has investigated its effectiveness among people with memory disorders. This review, thus, fills a gap in the literature by exploring the use of technology in supporting medication adherence among patients with memory issues by answering the following questions:

- (1) How can information technology support medication adherence among people with memory disorders?
- (2) What are the factors that impact the success of technology-based medication adherence interventions among these patients?

We define medication adherence in this article as the adherence to medication or care interventions prescribed by doctors or health care professionals.

Methods

This review was reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines. Our protocol was registered with the Open Science Framework on July 30, 2022.²⁴

Search Strategy

Five databases (Scopus, PubMed, Web of Science, ProQuest CENTRAL, IEEE Xplore) were searched for relevant publications published until March 15, 2023, when the search was conducted. Any study published prior to the date of search was included. The search strategy was developed based on the People, Intervention, Comparison, and Outcome (PICO) criteria.²⁵ The *population* was patients with memory disorders, the *outcome* was medication adherence, the *intervention* was technology-based, and the *comparison* was done by classifying the technology-based interventions based on their functions (reminder program, therapeutic education, treatment regimen simplification, or follow-up visits). For each of the PICO components, we associated a range of keywords that can help improve the inclusiveness of the searches. The search terms used are summarized in ►Fig. 1, with more details regarding the search terms used to find technology-based interventions summarized in ►Supplementary Appendix A (available online only). To select articles that have a population of people with memory issues, we followed a systematic method. We first identified review studies dealing with the same population,^{26,27} extracted the list of keywords they used, and then validated it with the librarian who updated and validated the list as they saw fit.

The search was conducted in a way to include any articles with any combination of these different search keywords in the title, abstract, keywords, or topic/summary (if available), depending on the database searched. For instance, the search was conducted in PubMed's title, abstract, and MeSH words. In ProQuest CENTRAL, the search was conducted in "all abstract & summary text." In Scopus, the search was

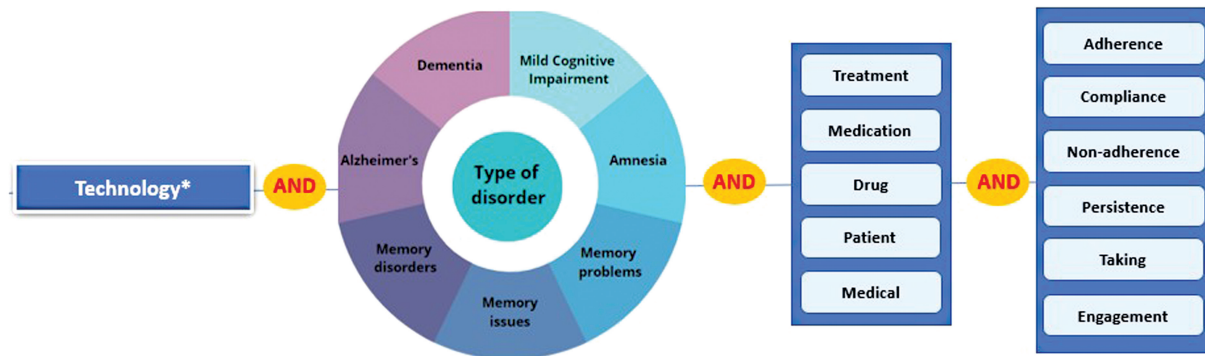


Fig. 1 Conceptual framework of the search terms used in the query of the studies for the review. *The words related to technology used in the search are attached in **►Supplementary Appendix A** (available online only).

conducted in article title, abstract, and keywords. In Web of Science, the search was conducted in “Topic,” which includes the title, abstract, and indexing of the papers. The search queries used are provided in **►Supplementary Appendix B** (available online only).

We exported the records retrieved to Endnote 20.1 (New Jersey, United States) for duplication removal and selection processes. It is noteworthy that some duplicates were found later in the cleaning process and were not captured initially by EndNote considering some differences in the titles and details across databases.

Selection Criteria

Inclusion Criteria

In this review we included studies that (1) are empirical (using surveys, interviews, experiments, and observational studies), (2) report outcomes related to the use of technology to support medication adherence, (3) conduct studies on people with memory disorders (Alzheimer's, amnesia, dementia, memory issues or disorders, or mild cognitive impairment [MCI]), (4) are peer-reviewed, and (5) are written in English.

Exclusion Criteria

We excluded any articles that (1) focus solely on healthy individuals or individuals with other health issues than memory disorders, (2) do not involve the use of technology-based interventions, (3) do not explore medication adherence issues, (4) are not peer-reviewed, or (5) that are not empirical. For instance, opinion papers, editorials, commentaries, and reviews were excluded.

Screening and Selection Process

We removed 58 duplicates from the studies (1.53% of the studies). We then downloaded the remaining 3,715 studies for title and abstract screening. This step resulted in the exclusion of 3,404 articles (90.21%) for noncompliance with the inclusion and exclusion criteria. Finally, a full review of the text for the remaining 311 articles was conducted. We excluded articles that were not relevant to the scope. Reviews, commentaries, and conceptual papers were also excluded. **►Fig. 2** outlines the PRISMA diagram illustrating

the review flow and the number of articles in each step. A data extraction form was used to extract standardized information from each paper.

Synthesis of the Results

Information regarding the population was reported as age and disease.^{28,29} To report the findings related to the role of technology in supporting medication adherence among the population selected, we used two frameworks adopted for this use case. We used the framework of Eicher et al³⁰ developed to classify the strategies adopted to improve medication adherence through technology applications. To screen for the factors impacting medication adherence, we used an adapted version of the conceptual framework developed by Peh et al and the World Health Organization's five dimensions of medication adherence.^{9,31} The review team discussed and synthesized information in an iterative process, considering the strengths and weaknesses of each conceptual model, as well as common factors and gaps across models in each patient group. **►Fig. 3** summarizes the conceptual model adopted in this study.

We define reminder programs as any electronic alerts communicated to patients or clinicians to remind them of an appointment, medication times, and doses designed to help alleviate the burden of manual reminders and improve scheduling and patient self-service. Therapeutic education tools refer to the technologies designed to teach patients how to adhere to medical processes through training or guides delivered electronically. All the phases of screening, labeling, and data extraction were conducted by the three reviewers. Any discrepancies between reviewers were resolved through discussion and consensus.

Quality Assessment and Risk of Bias

To ensure good quality of reporting, all the steps of selection were subject to consensus by all the authors. We also conducted a quality assessment of the papers following the Mixed Methods Appraisal Tool (MMAT).³² The MMAT assesses the quality of qualitative, quantitative, and mixed-methods studies.³² It focuses on methodological criteria and includes the nature of the study (randomized or nonrandomized clinical trial).³² The reviewers made a separate

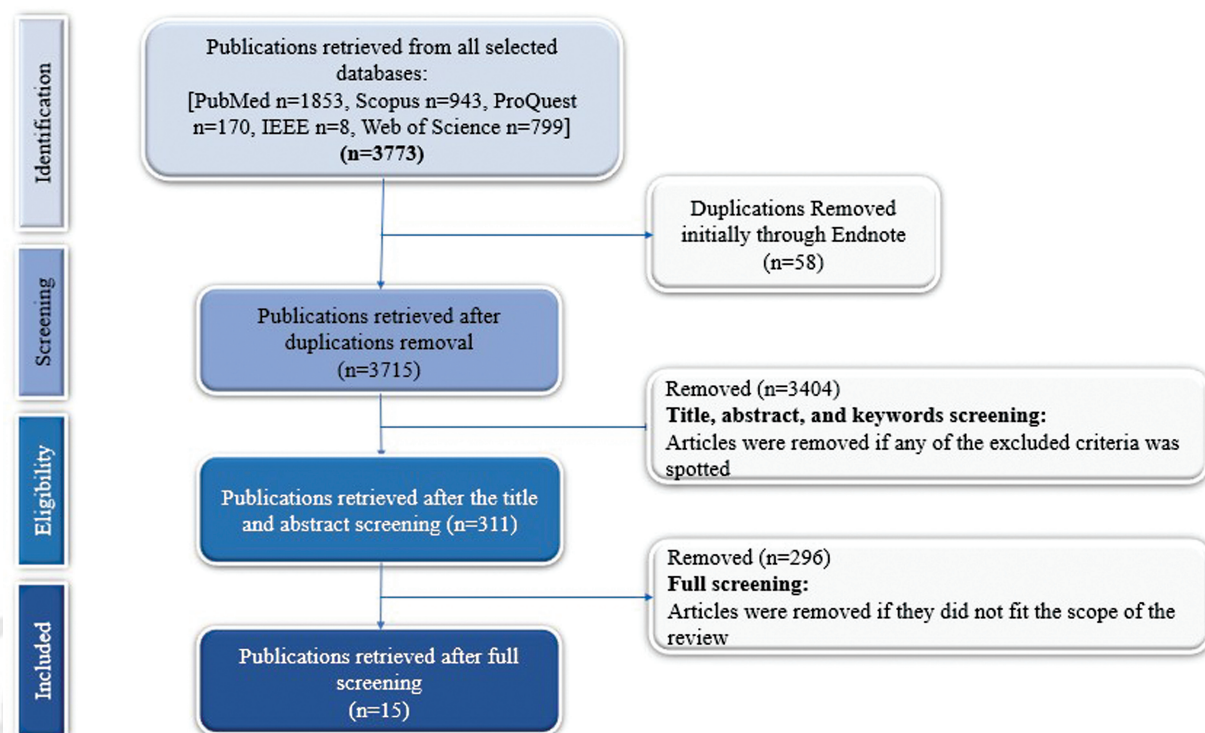


Fig. 2 PRISMA flow diagram of studies' selection.

judgment for each item (i.e., low risk of bias, high risk of bias, or unclear risk of bias). If a study fully met these criteria, the likelihood of various biases was low, and the quality grade was "A." If these criteria were partially met, the probability of bias was moderate, and the quality grade was "B." If these criteria were not met at all, the probability of bias was high, and the quality grade was "C." Articles with an overall quality level of A or B were included, and articles with an overall quality level of C were excluded.

Results

► **Fig. 2** illustrates the process of selection followed in this systematic review. After removing the duplicates and reading the full text, 15 studies were found to fit the scope of our study with consent from the authors. We used Endnote to manage the filtering process. Selected studies show that technology to support medication adherence started getting more attention in 2015. We used a systematic approach to extract information from the selected studies as the specific findings to report were agreed upon and for each of the articles, the author S.E. filled the columns after careful reading of the articles, then the other authors commented on the findings reported and enriched the reported outcomes. The findings are listed in ► **Table 1**. Only 33.33% of the included studies focused on both compliance and persistence of patients to the treatment suggested by doctors. In addition, 40% of them focused on supporting their compliance (duration/time wise) and 26.67% focused on the persistence of specific doses or movements.

Role of Technology in Supporting Medication Adherence among People with Memory Disorders

Therapeutic Patient Education

The role of technology in supporting medication adherence, particularly through effective therapeutic patient education, has emerged as a critical avenue in health care, with a specific focus on patients with MCI and dementia. Five studies examined the potential of therapeutic education enhanced by technology-based interventions to improve medication adherence. These studies have illuminated the substantial benefits that mobile technology (e.g., computer-based cognitive training [CCT], memory games, etc.) can offer in enhancing patients' commitment to their medical treatment plans. For instance, Turunen et al investigated the efficacy of CCT in older patients with dementia through the Finnish Geriatric Intervention Study to Prevent Cognitive Impairment and Disability (FINGER).³³ The authors showcased that patients who completed the training exhibited improved memory performance.³³ These findings underscore the positive impact of tailored interventions and memory training in educating patients on how to better adhere to the prescribed treatments.³³ This study also emphasizes the importance of considering the level of cognitive functioning when designing therapeutic interventions to boost adherence to CCT.³³ Adherence was measured through the number of sessions completed by patients. Participants who completed the training showed better memory performance.³³ Similarly, Gooding et al demonstrated that embedding CCT within a therapeutic milieu, BrainFitness, rather than as a standalone intervention,

Table 1 Findings of the study

Ref. Year	The objective of the study	Study design	Population supported	Type of technology	Name of the tool	How does technology impact adherence?	Factor impacting adherence	How was adherence evaluated?	Was the focus on compliance or persistence?	Findings	Limitations
42 2006	Using an electronic pillbox improves the quality of patient life and cognitive functions to maintain treatment adherence.	Experiment (N = 480 patients)	Dementia	Electronic pillbox	MedTracker pillbox	Reminder programs	-	Overall adherence/regimen adherence: percentage of taking medication within 1–2 hours before or after the prescribed times	Compliance and persistence	Using a 7-day electronic pillbox called the MedTracker, which records the time of day when a day's compartment is opened, a 5-week drug adherence trial using vitamin C supplements was conducted. Participants took this supplement from the MedTracker twice daily at prespecified times approximately 12 hours apart, ensuring that each person faced the same daily challenge. Improvement in patients' adherence to medication was observed.	The project is part of a bigger project for smart home monitoring systems. Physical activity was also tracked, not only medication adherence. But it was hard to track with the sensors because of the lack of spatial data (cannot know whether the user is in the kitchen or the living room)
43 2015	Exploring the role of game-based rehabilitation training in improving the patients' physical and cognitive performance and adherence to rehabilitation sessions	Experiment (N = 10 patients)	Dementia	Virtual reality-based training (VRT)	BrightBrainer	Simplifying treatment regimes	Patient-related factors	Cognitive performance, completion of tasks	Compliance	Integrative rehabilitation through games combining cognitive (memory, focusing, executive function) and physical (bimanual whole arm movement, grasping, task sequencing) elements are enjoyable for this population. Individual simulations adapted automatically to each participant's level of motor functioning to help him adhere to the training. The performance of patients using the system was better than the conventional physical rehabilitation sessions. The success of the intervention was related to the patients' need to remain physically and mentally fit; not only patients facing problems would want to use the tool.	Some limitations of this study include the relatively small sample size, lack of control participants, and the general severity of dementia
34 2016	Comparing patients' performance and engagement to treatment with cognitive training (online and traditional training)	Experiment (N = 74 patients)	Mild cognitive impairment (MCI), early Alzheimer's	Computer-based cognitive training (CCT)	Post Science's BrainFitness	Therapeutic patient education	Patient-related factor, condition-related factor, therapy-related factor	Performance on the mental state examination, verbal learning, verbal memory, adherence to the training	Compliance	Computerized cognitive training offers the most benefit when incorporated into a therapeutic milieu rather than administered alone, although both appear superior to more generic forms of cognitive stimulation. The patients' memory decline impacts their adherence to cognitive training. The engagement in a specific training depends on the type of training delivered (computerized or not)	The sample size was relatively small which may limit the statistical power of the results. The study did not include a control group that received no cognitive training which makes it hard to conclude the effects.

(Continued)

Table 1 (Continued)

Ref. Year	The objective of the study	Study design	Population supported	Type of technology	Name of the tool	How does technology impact adherence?	Factor impacting adherence	How was adherence evaluated?	Was the focus on compliance or persistence?	Findings	Limitations
46 2016	Exploring the role of health and wearable devices in promoting adherence to prescribed physical activity	Experiment (N = 21 patients)	Alzheimer's	Mobile Health (mHealth) technology-supported physical activity prescription	Zip, Fitbit Inc	Early follow-up visits and short-term treatment goals	Socio-economic factors	Adherence to physical activity (number of steps)	Persistence	The mobile health technology helped improve patients' adherence to physical activity as prescribed (more steps). The requirement of access to a wireless network may introduce socio-economic bias to the sample size.	Variability in measures was observed, but the sample is too small to assess potential reasons for the variability
35 2017	Improving patients' episodic memory and engagement with cognitive training through a gamified training	Experiment (N = 42 patients)	Amnesic mild cognitive impairment (aMCI)	Memory game on an iPad for cognitive training	Game Show	Therapeutic patient education	Patient-related factor	Episodic memory and motivation to continue the training	Compliance	The memory game helped improve the patients' adherence to treatment plans by improving their enjoyment and cognition to impact their motivation to continue after each hour. Factor impacting the adherence was: motivation.	Larger, more controlled trials are needed to replicate and extend these findings.
33 2019	Investigating older adults' adherence to a long-term computerized cognitive training program and determining participant characteristics associated with adherence to the training.	Experiment (N = 631 patients)	Dementia	CCT	FINGER initiative	Therapeutic patient education	Condition-related factor, patient-related factor	The number of completed CCT sessions	Persistence	CCTs can help improve older patients' adherence to training. Encouraging computer use and considering the level of cognitive functioning may help boost adherence to CCT. Previous computer use was the only factor associated with more training sessions completed.	Technology difficulty of use among elderly people can impact the study outcomes
47	Improving patients' adherence to treatment. Improves their working memory, delayed memory, and cognitive function.	Experiment (N = 66 patients)	MCI	Virtual interactive working memory training (VIMT)	CogniPlus	Early follow-up visits and short-term treatment goals	Condition-related factor	Working memory, number of completed sessions	Compliance	Memory deterioration impacts the patients' ability to adhere to treatment plans	The study used a single-blind design, which may introduce bias as the participants were aware of their group assignment.
41 2019	Exploring the impact of using an automatic medication dispenser on patients' adherence to medication.	Experiment (N = 4 patients)	Alzheimer's	Automatic medication dispenser (AMD)	Pivotell	Reminder programs	-	Medication adherence: medications remaining during 1 week	Compliance	The tool improved medication adherence throughout the years (3–4.5).	Very small sample size and not all subjects adhered to the intervention.
40 2020	Assessing the effectiveness of employing the MeMo (Memory Motivation) Web app concerning cognitive and behavioral symptoms in patients with neurocognitive disorders.	Experiment (N = 46 patients)	Neurocognitive disorders	Web-app	MeMo	Reminder programs	Health care systems-related factors	Motivation, attention, completed tasks	Compliance and persistence	Reminding patients about regular use at the moment of prescription and possibly the requirement of regular follow-ups to check prescription adherence are important in deciding their compliance with the planned treatment. The web app has two parts: memory and mental flexibility/attention. It helped improve the regular use of the training by patients, their attention, and their motivation. It has cognitive and behavioral efficacies.	Short-term interventions cannot inform as well as regular-use interventions.

Table 1 (Continued)

Ref. Year	The objective of the study	Study design	Population supported	Type of technology	Name of the tool	How does technology impact adherence?	Factor impacting adherence	How was adherence evaluated?	Was the focus on compliance or persistence?	Findings	Limitations
36 2020	Exploring the role of gamification in enhancing patients' adherence to cognitive training	Experiment (N = 41 patients)	MCI, Parkinson's disease	Game-based cognitive training (GCT)	AquaSnap	Therapeutic patient education	Patient-related factors	Medication adherence, cognitive assessment questionnaire, adherence to training	Compliance and persistence	The technology used is an adaptive cognitive training game exercising five cognitive domains: attention, working memory, episodic memory, psychomotor speed, and executive function. Various game elements are incorporated to promote adherence, such as goals, challenges/missions, reward systems, personalization, and 3D environments. To assess the feasibility of stand-alone GCT, patients were trained unsupervised at home, and only reactive support was provided when requested by the patient. One of the factors impacting patients' adherence to treatment is the need to travel to an institute for the assessments.	The usability and feasibility of the gamified interventions should be investigated to generalize the findings. Longer follow-up periods are needed. Larger studies with robust cognitive outcomes that also focus on translational aspects of daily life are needed to determine the effectiveness of this intervention.
39 2020	Exploring the impact of medication ambient display system on medication adherence.	Experiment (N = 16 patients)	MCI	Patient Portal	Medication Ambient Display (MAD)	Reminder programs	Therapist-related factors	Pills counting, MAQ (Medication Adherence Questionnaire)		The system uses different abstract modalities to provide external cues that enable older adults to easily take their medications on time and be aware of their medication adherence. Some participants face problems managing their medications, such as accumulating and confusing medications because they look alike.	The use of financial incentives has been questioned because they may provide inducements to participate in a study for financial purposes only, and vulnerable populations are prone to be enticed by the financial reward and be more willing to accept any study risks
37 2021	Exploring the feasibility of the tool as a real-time potential balance training and falls prevention intervention for older adults with MCI.	Experiment (N = 15 patients)	Mild cognitive impairment	Online falls prevention intervention	Tai ji Quan program	Exploring the feasibility of the tool as a real-time potential balance training and falls prevention intervention for older adults with MCI.	Patient-related factors, therapy-related factors	Attendance rate, number of falls	Compliance	The patients showed compliance and potential benefit from the at-home virtual interactive program delivered in real-time to improve balance training. Older adults with minimal computer or smartphone experience may feel uncomfortable accessing this new exercise platform, which requires completing various logistical tasks, ranging from tracking the online (Zoom) class link, signing on to a session, and positioning their device(s) for better viewing of the instructor (or vice versa) to	People without access to technology resources would not be able to participate in technology-driven virtual exercise interventions.

(Continued)

Table 1 (Continued)

Ref. Year	The objective of the study	Study design	Population supported	Type of technology	Name of the tool	How does technology impact adherence?	Factor impacting adherence	How was adherence evaluated?	Was the focus on compliance or persistence?	Findings	Limitations
38 2021	Examining how a technology system can help augment traditional rehabilitation strategies for patients by improving engagement in therapy sessions and achieving better functional outcomes	Experiment (N = 48 patients)	Dementia	Patient portal	It's Never 2 Late (IN2L)	Examining how a technology system can help augment traditional rehabilitation strategies for patients by improving engagement in therapy sessions and achieving better functional outcomes	Patient-related factors	Pittsburgh rehabilitation participation scale (PRPS) for rehabilitation engagement, goal attainment	Persistence	N2L technology was able to increase treatment engagement and enhance rehabilitation outcomes among older adults with dementia. The time issues were impacting patients' adherence to the plans.	Limited sample size
44 2022	Improving patients' spatial cognition and hippocampal function using virtual reality training program that uses active navigation instead of passive navigation.	Experiment (N = 56 patients)	MCI	Virtual reality-based spatial cognitive training (VRST)	VR-SCT	Improving patients' spatial cognition and hippocampal function using virtual reality training program that uses active navigation instead of passive navigation.	Condition-related factor	Completion of sessions	Compliance and persistence	To improve patients' adherence to treatment, this training aims to improve their hippocampal function (spatial cognition and episodic memory)	Limited sample size
45 2022	Describing older adults' engagement with an eHealth intervention, identifying factors associated with engagement, and examining associations between engagement and changes in dementia risk factors	Experiment (N = 1389 patients)	Cognitive decline	eHealth	Healthy Aging Through Internet Counselling in the Elderly (HATICE)	Describing older adults' engagement with an eHealth intervention, identifying factors associated with engagement, and examining associations between engagement and changes in dementia risk factors	Patient-related factors, therapies-related factors	Engagement in eHealth interventions; log-in frequency, number of lifestyle goals set, measurements entered, and messages sent to coaches, and percentage of education materials read.	Persistence	Limited computer experience, lower level of education, and poorer cognition negatively impacted patients' engagement in the plans. Greater engagement with an eHealth lifestyle intervention was associated with greater improvement in risk factors in older adults.	Limited computer experience, lower level of education and poorer cognition negatively impacted patients' engagement in the plans.

fostered better adherence to cognitive rehabilitation exercises among patients with MCI and Alzheimer's suggesting the importance of the therapeutic milieu in the success of such technologies in educating patients.³⁴

Furthermore, the integration of gamified experiences, as seen in studies such as Savulich et al, demonstrated the power of motivation in promoting adherence through memory games on iPads.³⁵ This study highlighted the importance of therapeutic education of patients through Game Shows in supporting their engagement in cognitive training.³⁵ The study found that the main factor impacting adherence is motivation, which suggests designing therapeutic education and training sessions that can motivate patients with memory problems continuously to engage and improve their enjoyment and cognition.³⁵ The incorporation of game-based cognitive training mobile tools like AquaSnap further emphasizes the significance of integrating personalized elements such as goals, rewards, and challenges in therapeutic education interventions to foster adherence.³⁶ Finally, online interventions such as Tai Ji Quan showcased how technology could offer multifaceted solutions, as Li et al employed an online platform to ensure patients' commitment to balance training and fall prevention, leveraging both education and interactive engagement.³⁷

Reminder Programs

Five studies harnessed the potential of technology to bolster adherence through innovative reminder programs, strategically designed to bolster patients' adherence to treatments. These interventions leveraged a variety of platforms, including patient portals, web applications, and innovative solutions for medication consumption, such as automatic medication dispensers (AMDs) and electronic pillboxes. For instance, Zarit et al ingeniously employed the patient portal "It's Never 2 Late" to foster adherence among patients with dementia, ensuring their commitment to rehabilitation exercises and therapy sessions by providing them with continuous reminders from doctors on the importance of the interventions and facilitating their communication.³⁸

The integration of medication ambient display systems, shared via patient portals, ingeniously employed various abstract modalities. This approach provided external cues that proved invaluable for older adults with MCI, facilitating the timely intake of medications while enhancing awareness of adherence to prescribed regimens.³⁹ In the same vein, technology-driven medication consumption tools, exemplified by the web-app MeMo (Memory Motivation), embraced the potential of health information technology. By engaging patients with neurocognitive disorders in regular reminders about their prescriptions and emphasizing the significance of compliance, these interventions emerged as powerful agents of adherence support through continuous reminder programs.⁴⁰ The transformative role of health information technology was further underscored by Kamimura, who introduced patients to an AMD named Pivotell. Over a span of 3 to 4.5 years, Pivotell emerged as an agent of change by significantly enhancing medication adherence among patients grappling with Alzheimer's.⁴¹ The pinnacle of this

technology-driven approach to adherence enhancement was epitomized by the electronic pillbox, MedTracker.⁴² This advanced tool, equipped with the ability to record the precise time of day when each compartment was opened, ensured a sophisticated level of monitoring. Hayes et al's drug adherence trial, encompassing vitamin C supplements, highlighted how MedTracker's strategic design facilitated adherence improvements.⁴² By orchestrating a regimen wherein participants took the supplement twice daily, precisely 12 hours apart, MedTracker not only introduced consistency but also embraced health information technology to standardize the daily challenge across all individuals.⁴²

Simplifying Treatment Regimens

Three innovative interventions prioritized the simplification of treatment regimens as a means to enhance patients' adherence. Among these interventions, two harnessed the potential of virtual reality (VR), while the third leveraged an electronic health tool (eHealth). The integration of VR training (VRT), exemplified by platforms like BrightBrainer, ingeniously merged cognitive and physical elements within interactive games. This approach not only encouraged patients with dementia to actively engage in rehabilitation training but also facilitated adherence regimens through immersive experiences.⁴³ Additionally, a distinct spatial cognitive VRT tool emerged as a potent strategy to enhance adherence. This tool effectively targeted patients with MCI, aiming to enhance their hippocampal function encompassing spatial cognition and episodic memory.⁴⁴ By aligning treatment regimens with spatial training, this intervention embraced technology to optimize adherence and facilitate it among individuals facing cognitive challenges.⁴⁴ Furthermore, the advent of technology-driven eHealth initiatives ushered in a paradigm shift in supporting adherence by simplifying treatment regimens.⁴⁵ "Healthy Ageing Through Internet Counselling in the Elderly" (HATICE), for instance, capitalized on the power of eHealth to provide comprehensive support to elderly patients with cognitive decline.⁴⁵ By facilitating personalized training, empowering patients to achieve lifestyle objectives, sharing educational resources, and facilitating consistent communication with coaches, this initiative symbolized the transformational role of technology in streamlining treatment regimens for optimized adherence.⁴⁵ Collectively, these interventions underscore the pivotal role of technology in advancing patient-centric approaches, ultimately fostering adherence by strategically simplifying treatment regimens.

Early Follow-Up Visits

The transformative potential of technology was prominently evident in the realm of medication adherence, particularly concerning patients grappling with Alzheimer's. mHealth and wearable devices, exemplified by Zip from Fitbit Inc., emerged as remarkable allies in promoting adherence to prescribed physical activity.⁴⁶ This technological synergy facilitated seamless remote monitoring, allowing health care providers to remotely track changes in patients' weekly step counts, an innovative approach that revolutionized

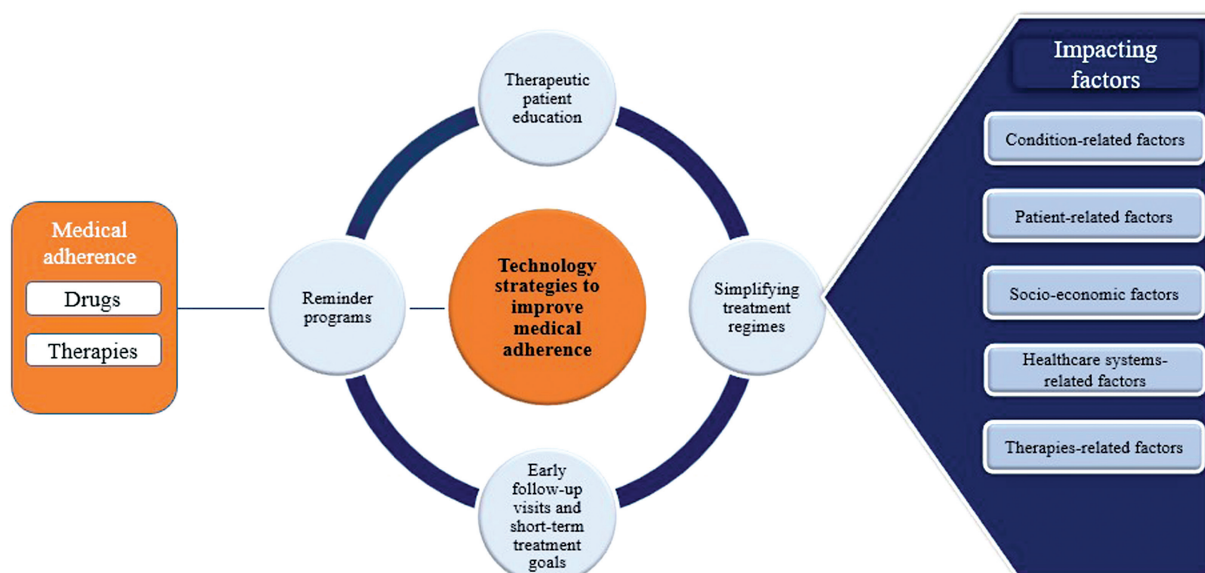


Fig. 3 Conceptual model of technology's role in supporting medication adherence and the factors impacting that.

follow-up practices.⁴⁶ Furthermore, the dawn of virtual interactive working memory training (VIMT) showcased technology's indispensable role in supporting medication adherence among elderly patients with MCI.⁴⁷ Yang et al illuminated how the implementation of the CogniPlus VIMT fostered memory training. This visionary program not only empowered patients to uphold their working memory but also remarkably curtailed the pace of cognitive deterioration. The hallmark of this approach lies in its continuous professional oversight, where technology served as the conduit for ongoing follow-up visits, ensuring sustained medication adherence.⁴⁷ In essence, these findings underscored how technology, particularly through facilitated early follow-up visits, stands as an instrumental force in reinforcing medication adherence among patients battling cognitive challenges.

Factors Challenging Technology-Based Medication Adherence Interventions among People with Memory Disorders

Based on our study's findings, different factors impact medication adherence among patients with memory disorders.

Human-Computer Interaction

Several key factors influencing medication adherence were closely intertwined with the patients themselves, underlining the significance of information technology in facilitating their commitment to treatment plans. Despite a genuine need for treatment, patients' dedication to maintaining mental and physical well-being emerged as a pivotal determinant of adherence.⁴³ Their motivation to persist with specialized technology-based training substantially impacted their engagement throughout the duration of interventions.³⁵ Moreover, the ease of technology utilization and user experience stood out as critical elements shaping intervention effectiveness.^{33,34,37} Intriguingly, prior familiarity with computer use often emerged as a pivotal factor influencing the completion of training

sessions.³³ Additionally, low education levels, memory decline, and diminished cognitive function had negative influences on patients' active involvement in the technology-based intervention plans.^{33,34,45,47} Some individuals faced auditory impairments due to aging, further complicating communication with trainers and health care providers through online tools, thereby hindering engagement in similar treatment interventions.³⁷ The perpetual challenge of time scarcity further resonated as an impediment to adherence, given the extensive commitments associated with prescribed technology-based training regimens despite its flexibility.³⁸

Challenges Related to Technology Integration in Health Care Systems and Interventions' Design

On another tangent, several adherence-related factors were rooted not in the patients but rather in the integration of technologies in therapy design and in the health care systems' frameworks, spotlighting the transformative role of information technology. Foremost, the necessity for regular follow-ups to enhance compliance underscored the potential of technology-enabled remote monitoring and communication, although not all interventions seamlessly provided this advantage.⁴⁰ Adherence also pivoted on the delivery mode of training, whether online or in-person and the structure of interventions, influencing their effectiveness.³⁴ Particularly, interventions yielded greater benefits when nested within a therapeutic environment, emphasizing technology's potential to foster an augmented, synergistic approach to adherence enhancement.³⁴ Some interventions involved intricate tasks and steps that not all patients found comfortable, ranging from online session sign-ins to tracking links and positioning devices, underlining the importance of user-friendly technology interfaces and interventions.³⁷ Additionally, advanced technology in some interventions inadvertently acted as a barrier, impeding user progress and highlighting the necessity for seamless user experiences.⁴⁵ Unequal access to care due to varied internet

connectivity underscored the importance of equitable technological provisions for all patients.^{36,46} The time-intensive nature of medication adherence-supporting interventions, culminating from complex health care regimens, emerged as a consistent challenge, thereby emphasizing the challenge that technology can face in managing and streamlining adherence activities.³⁸ Lastly, the conundrum of medication accumulation and confusing labels can make it hard for technology-based intervention designers to simulate reality and support patients' adherence.³⁹

Risk of Bias and Quality Assessment

► **Supplementary Appendix C** (available online only) and ► **Fig. 4** summarize the quality assessment results run by the three authors. All articles reported adequate random sequence generation and thus had a low risk of selection bias. Methodological quality, as appraised by MMAT for the included studies, ranged from 80 to 100% with eight studies rated 100%. The studies overall did not have a high risk of bias. The data collection methods in the studies chosen were found to adequately answer the research questions, and the data found addressed the objectives. The interpretation of the data was consistent with the methods, and confounding factors were taken into consideration.

As shown in ► **Fig. 4**, no study had any detection bias related to binding outcomes assessment, or attrition bias related to providing complete outcome data. Also, all studies provided comparison at baseline with control groups. However, three studies had selection bias (random sequence generation),^{34,38,41} and some participants in four studies dropped out of the intervention (performance bias).^{33,36,37,47}

Discussion

The present study systematically reviewed and analyzed the role of technology in supporting medication adherence among individuals with memory disorders. The results indicate a growing interest in the field since 2015, suggesting an increasing recognition of the potential of this technology in enhancing the adherence of these patients. Studies investigating patients' adherence focus only on the medication side.^{48–50} This is, to our knowledge, the first review investigating the technology-based interventions that support medication adherence among people with memory disorders. ► **Fig. 5** highlights the most important recommendations resulting from the review findings.

The findings of the 15 included studies have elucidated the potential of technology in supporting medication adherence among patients with memory disorders. The selected studies underscore the multifaceted ways in which technology can enhance patients' commitment to treatment plans, thereby addressing the intricate challenges posed by memory impairments. The exploration of therapeutic patient education, reminder programs, simplifying treatment regimens, and early follow-up visits through technology-driven interventions highlights the diverse avenues through which technology can contribute to fostering medication adherence. Patients with memory issues require more assistance to meet their medication adherence targets, as a decline in memory and cognitive functions interferes with their ability to successfully perform a routine of daily living and instrumental activities.⁵¹ Many technologies have shown reliable results in supporting medication adherence among different

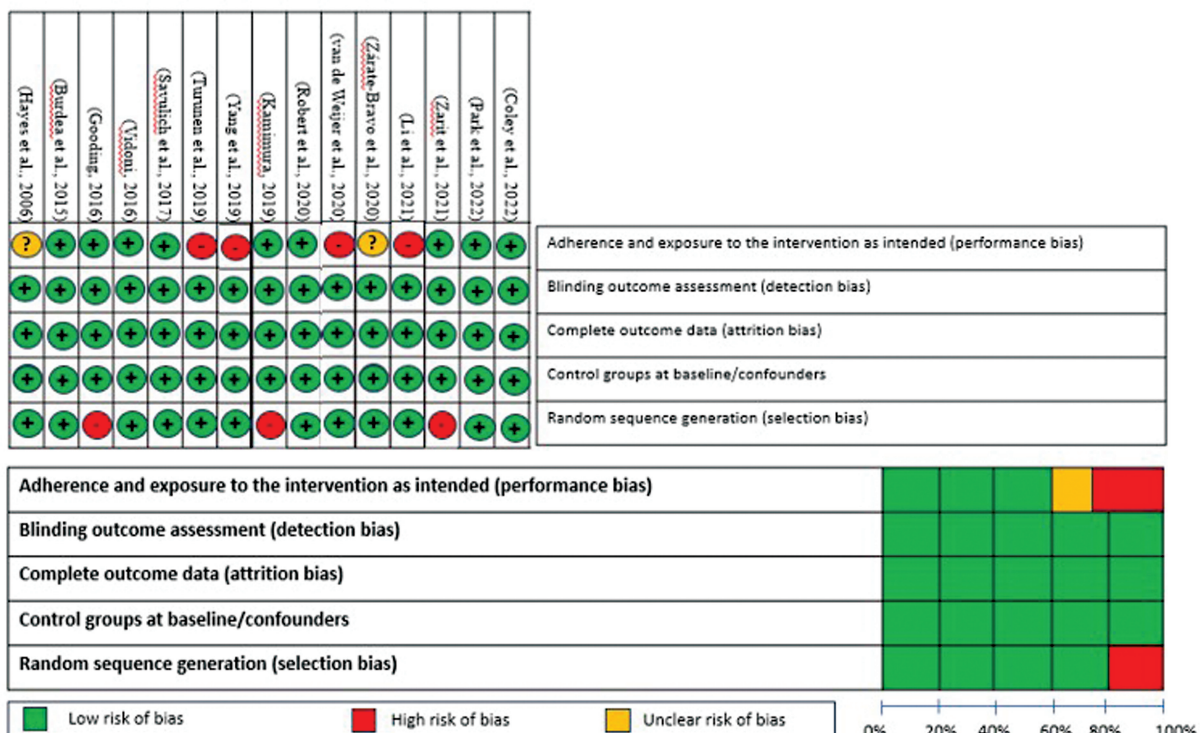


Fig. 4 Risk of bias summary for the included studies.

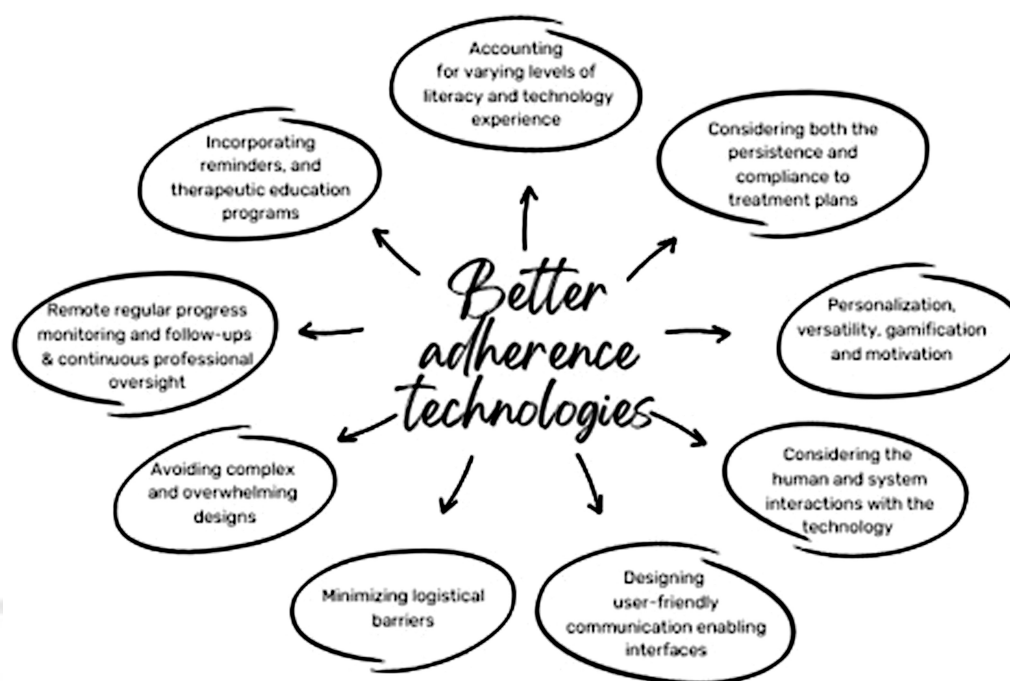


Fig. 5 Technology improvement suggestions derived from the study's findings.

populations,^{52,53} and especially people with memory disorders as shown through our review findings. Through the discussion of the findings of this review, we wanted to highlight important recommendations that can help in the design of future technology-based adherence programs.

Therapeutic Patient Education

These technologies were shown by our review to be able to support medication adherence through different strategies such as supporting therapeutic patient education.^{33–35} These findings coincide with other findings showing that shaping the patients' beliefs related to medications and educating them can strongly influence the patients' adherence to treatment.^{54,55} Thus, we conclude that *technologies should be able to educate patients on how and why they should adhere to treatments to raise their awareness of the importance of complying with their medication and treatment regimens*. In addition, the findings showed that *personalization, gamification, and motivation* were factors that impacted the success of the used technologies.^{33–36} First, tailoring the interventions to patients' cognitive abilities can contribute to the success of the technology used.³⁶ This finding suggests that technologies should be *personalized to the cognitive functioning of the patient by incorporating adaptive features that assess patients' abilities and adjust the educational content accordingly to optimize the outcomes*. Second, we found that gamified experiences can enhance motivation among patients with memory issues to adhere to the technology-based adherence interventions.³⁵ This finding coincides with other studies showing that gamification features can improve patients' adherence to mental well-being interventions.⁵⁶ We thus conclude that technology developers should consider *integrating gamification elements like rewards, challenges, and engaging interfaces into therapeutic interventions to maintain patient motivation*.

Considering the importance of motivation in the adherence of our patients to the interventions as shown by Savulich et al,³⁵ technologies used for adherence support should focus on *incorporating interactive elements and progress-tracking features to ensure sustained engagement and enjoyment*.

Reminder Programs

Studies have shown that reminder-based interventions have the potential to improve adherence to medications among patients with chronic conditions.⁵⁷ This can explain why technology-based interventions were found to support adherence among patients with memory issues in our review.^{33,46} With the deterioration of memory capabilities among vulnerable populations, reminder programs can help them cope with their daily lives and adapt to their physiological changes.^{58,59} Thus, adherence technologies should consider *incorporating reminders to assist patients with their medical and treatment goal achievement*. In addition, the findings of this review showed that *user-friendly interfaces that facilitate communication and the versatility aspects of the technologies can help in its success in supporting adherence, which should be considered by technology designers and developers*. Technologies should facilitate timely reminders and easy-to-access intuitive updates that build a bridge of communication between health care providers and patients with memory issues. Furthermore, as captured by our findings, patients have different levels of technology familiarity which would impact their adherence to the interventions.³³ Thus, reminder systems should have *clear and simple interfaces that can accommodate various technological ease-of-use levels*. Lastly, considering that the different solutions offered in this review were shown to be beneficial, it is important to notice that technology-driven medication adherence tools should be adaptable to the preferences of patients by making

them available on multiple devices and different formats accommodating the different user needs.

Early Follow-Up Visits

Through the different technologies used, this review underscored the importance of technology in supporting medication adherence among patients with memory problems through early follow-up visits.^{46,47} These studies showed that by enabling remote monitoring accountability and consistent follow-ups, this technological synergy facilitated medication adherence among patients with memory loss.^{46,47} *Maintaining consistent and timely follow-up components should be considered by designers of similar technologies.* The technology should provide *accurate data on patients' activities and ensure a seamless experience for both patients and health care providers.* In addition, VIMT was successful because it incorporated *ongoing professional oversight.*^{46,47} Thus, technology designers should *consider enabling not only remote progress monitoring but also the possibility to intervene and provide support when needed.*

Challenges of Medication Adherence Technology-Based Interventions Success

The findings of this study related to the challenges of the technology-based interventions success shed light on various dimensions (other than the technology design) that may impact the effectiveness and feasibility of these interventions. These factors highlight the important interplay between technology design, patient acceptance, and implementation within the larger health care context, underscoring the need for a *comprehensive approach* to addressing adherence challenges. For instance, we found that the successful human-computer interaction experience was crucial to the success or failure of some of the interventions.⁴³ Many studies have shown that difficulty in technology use can limit the success of interventions in health care settings which not only impact small-scale organizations but also cover large-scale clinical implementations.⁶⁰ Thus, future large-scale interventions should consider addressing difficulty challenges to ensure a higher probability of the interventions.

Despite the memory disorders, patients who were motivated to maintain their well-being tended to exhibit better adherence results.³⁵ We thus conclude that for the technology to succeed in enhancing adherence, it should *align with the patients' motivations and goals.* Furthermore, user experience and ease of use emerged as pivotal factors as patients with technology familiarity showed better adherence and completion rates while lower educational levels and cognitive impairments hindered active engagement.³⁵ Thus, the design of technology-based interventions in small- and large-scale implementation settings should *account for users with varying levels of technology literacy and cognitive abilities and test the usability of the technologies to ensure iterative improvements in the design based on the users' feedback.* Additionally, as shown in this review, as important as technology-enabled remote monitoring can be, adherence interventions should not only be technology-oriented but *proper supervision and oversight by health care professionals*

is needed on a regular basis to ensure consistent communication between doctors and their patients. Moreover, we found that the mode of delivering the intervention (in-person or online) and the overall structure of the interventions influenced their effectiveness as one of the studies showed that embedding interventions within a therapeutic milieu yields better results.³⁴ This finding suggests that technology-based adherence interventions *should be designed in a way that minimizes the logistical barriers that may occur preventing patients from continuing to adhere.* Considering the time constraints reported by some of our patients, the designers of technology-based interventions should carefully consider their complexity to avoid overwhelming patients. Finally, considering both *compliance and persistence* is pivotal when crafting interventions that aim to enhance adherence among patients with memory issues. These factors collectively ensure the proper execution of treatment plans and the sustained commitment to them. Neglecting either element could undermine the intervention's effectiveness and hinder positive health outcomes. By encompassing both compliance and persistence in intervention design, health care professionals can create holistic strategies that maximize treatment impact and foster lasting improvements for individuals facing memory-related challenges.

Limitations

This study is, to our knowledge, among the first studies to acknowledge the importance of using technology to support medication adherence among patients with memory disorders. However, it has some limitations that are worth acknowledging. First, despite the importance of the results in some studies, findings cannot be generalized because of the sample size. Also, only studies in the English literature were included as done in previous studies. However, we acknowledge that because of this restriction, we may have missed potential studies with important findings. In addition, some studies do not account for demographics, implying that some groups of people may not be well represented. Also, the chosen keywords employed in this systematic review were thoughtfully selected to comprehensively cover the relevant literature within the scope of our research question. However, it is important to acknowledge that despite our best efforts, it is possible that some pertinent articles may have been inadvertently excluded. We recognize that systematic reviews strive for completeness, and while our approach was designed to capture a wide range of relevant studies, there may still be valuable contributions that lie beyond the boundaries of our chosen keywords. Lastly, some studies have a limited sample size, which makes it hard to generalize the results to broader populations. Additionally, it's important to note that the generalizability of certain studies may be limited due to variations in their quality assessment scores. This limitation is particularly evident in feasibility and exploratory studies, as well as in some randomized control trials where participants did not fully adhere to the intervention throughout the study duration. Consequently, to arrive at more robust and widely applicable conclusions, additional evidence is required. Another point to consider is that the few patients who did

not adhere to the interventions were not asked about the reasons behind dropping out of the experiments. It is noteworthy that nonadherence to the intervention may be related to the fact that technology use added extra burden on the participants who already are suffering from the medication adherence burden. Future studies should investigate the burden bias related to the different interventions on medication adherence among patients with memory disorders.

Conclusion

This systematic review addressed an important gap in the area of medication adherence. It focuses on the role of health information technology in supporting medication adherence among patients with memory disorders. It also captures the factors that impact patients' medication adherence in the selected studies. This study adds to the literature by summarizing the main four areas where technology can contribute to medication adherence support: therapeutic education, reminder programs, simplifying treatment regimens, and early follow-up visits. The factors that impacted patients' adherence to technology-based treatment and medication were clustered into human-computer interaction challenges and system-computer integration challenges.

Q4

Highlights

- (1) Health information technology solutions have the potential to support medical adherence among patients with memory disorders.
- (2) The available technologies fall under four categories based on their impact: therapeutic patient education, simplifying treatment regimens, early follow-up visit and short-term treatment goals, and reminder programs.
- (3) Technology-based adherence interventions are impacted by human-computer interaction factors and technology-system integration factors.

Clinical Relevance Statement

Medication adherence is a real challenge for people who suffer from memory disorders. This review aims to show the potential that health information technology has in supporting medication adherence among this population. The findings shed light on the factors that should be considered when designing a technology that aims to support medication adherence.

Multiple Choice Questions

1. What is medication adherence?
 - a. Ability to adhere to a treatment plan prescribed by a health care professional.
 - b. Health information technology
 - c. Patient-centered initiative
 - d. Compliance to laws

Correct Answer: The correct answer is option a. Medication adherence is defined by the World Health Organization as

“the degree to which the person's behavior corresponds with the agreed recommendations from a health care provider.” <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3191684/>.

2. Who has memory disorder?
 - a. People with dementia, Alzheimer's, etc.
 - b. Type of technology
 - c. Health system environment
 - d. Cardiovascular problem.

Correct Answer: The correct answer is option a. Memory disorders occur when damage to certain parts of the brain prevents or reduces the ability to store, retain, or remember memories. <https://www.premierhealth.com/services/neuroscience/memory-disorders#:~:text=Memory%20disorders%20occur%20when%20damage,ability%20to%20perform%20simple%20tasks.>

Protection of Human and Animal Subjects

No human subjects were involved in the project.

Conflict of Interest

None declared.

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THIEME

Supplementary Appendix A Technology Mesh Words Used

Informatics OR technology OR "Information Technology" OR "Electronic Health Records" OR "Software" OR Telemedicine OR informatics OR "information technologies" OR groupware OR "group ware" OR "computer supported cooperative work" OR "computer-supported cooperative work" OR "shared workspace" OR "shared work-space" OR "shared work space" OR "information and communication technology" OR "information & communication technology" OR "information and communication technologies" OR "information & communication technologies" OR "information systems" OR "information system" OR "data system" OR "data systems" OR "informational system" OR "computer system" OR "computer systems" OR "web-based" OR "internet-based" OR "web 2.0" OR "world wide web" OR "world-wide web" OR "user-computer" OR "human-computer" OR "computer interface" OR "computer interfaces" OR "health information exchange" OR "health information exchanges" OR "interoperability" OR "interoperable" OR "electronic health record" OR "electronic health records" OR "electronic medical record" OR "electronic medical records" OR "electronic patient record" OR "electronic patient records" OR "patient portal" OR "patient portals" OR "personal health record" OR "personal health records" OR "personal-health record" OR "personal-health records" OR "tethered health record" OR "tethered health records" OR mhealth OR "m-health" OR "mobile health" OR "touch screen" OR touchscreen OR "smart phone" OR ehealth OR "e-health" OR "electronic health" OR "electronic intervention" OR "digital intervention" OR "electronic tool" OR "digital tool" OR "digital technology" OR "electronic technology" OR "digital technologies" OR "digital health" OR "electronic technologies" OR "e-communication" OR "e-intervention" OR telecounseling OR telemonitoring OR telemedicine OR "video visit" OR "video consult" OR "video consultation" OR telehealth OR "remote monitoring" OR "internet communication technologies" OR "internet communication tool" OR "clinical decision support" OR teleconsultation OR teleconsultations.

Supplementary Appendix B

Search query for PubMed

((((Informatics OR technology OR "Information Technology" OR "Electronic Health Records" OR "Software" OR Telemedicine OR informatics OR "information technologies" OR groupware OR "group ware" OR "computer supported cooperative work" OR "computer-supported cooperative work" OR "shared workspace" OR "shared work-space" OR "shared work space" OR "information and communication technology" OR "information & communication technology" OR "information and communication technologies" OR "information & communication technologies" OR "information systems" OR "information system" OR "data system" OR "data systems" OR "informational system" OR "computer system" OR "computer systems" OR "web-based" OR "internet-based" OR "web 2.0" OR "world wide web" OR "world-wide web" OR "user-computer" OR "human-computer" OR "computer interface" OR "computer interfaces" OR "health information exchange" OR "health information exchanges" OR "interoperability" OR "interoperable" OR "electronic health record" OR "electronic health records" OR "electronic medical record" OR "electronic medical records" OR "electronic patient record" OR "electronic patient records" OR "patient portal" OR "patient portals" OR "personal health record" OR "personal health records" OR "personal-health record" OR "personal-health records" OR "tethered health record" OR "tethered health records" OR mhealth OR "m-health" OR "mobile health" OR "touch screen" OR touchscreen OR "smart phone" OR ehealth OR "e-health" OR "electronic health" OR "electronic intervention" OR "digital intervention" OR "electronic tool" OR "digital tool" OR "digital technology" OR "electronic technology" OR "digital technologies" OR "digital health" OR "electronic technologies" OR "e-communication" OR "e-intervention" OR telecounseling OR telemonitoring OR telemedicine OR "video visit" OR "video consult" OR "video consultation" OR telehealth OR "remote monitoring" OR "internet communication technologies" OR "internet communication tool" OR "clinical decision support" OR teleconsultation OR teleconsultations) AND (amnesia OR dementia OR "mild cognitive impairment" OR alzheimer OR "memory issue" OR "memory problem" OR "memory disorder"))) AND (treatment OR medication OR drug OR patient OR medical)) AND (adherence OR compliance OR non-adherence OR persistence OR taking OR engagement))) AND (((amnesia OR dementia OR "mild cognitive impairment" OR alzheimer OR "memory issue" OR "memory problem" OR "memory disorder"[MeSH Terms]) AND (treatment OR medication OR drug OR patient OR medical [MeSH Terms])) AND (adherence OR compliance OR non-adherence OR persistence OR taking OR engagement[MeSH Terms])) AND (Informatics OR technology OR "Information Technology" OR "Electronic Health Records" OR "Software" OR Telemedicine OR informatics OR "information technologies" OR groupware OR "group ware" OR "computer supported cooperative work" OR "computer-supported cooperative work" OR "shared workspace" OR "shared work-space" OR "shared work space" OR "information and communication technology" OR "information & communication technology" OR "information and communication technologies" OR "information & communication technologies" OR "information systems" OR "information system" OR "data system" OR "data systems" OR "informational system" OR "computer system" OR "computer systems" OR "web-based" OR "internet-based" OR "web 2.0" OR "world wide web" OR "world-wide web" OR "user-computer" OR "human-computer" OR "computer interface" OR "computer interfaces" OR "health information exchange" OR "health information exchanges" OR "interoperability" OR "interoperable" OR "electronic health record" OR "electronic health records" OR "electronic medical record" OR "electronic medical records" OR "electronic patient record" OR "electronic patient records"

OR "patient portal" OR "patient portals" OR "personal health record" OR "personal health records" OR "personal-health record" OR "personal-health records" OR "tethered health record" OR "tethered health records" OR mhealth OR "m-health" OR "mobile health" OR "touch screen" OR touchscreen OR "smart phone" OR ehealth OR "e-health" OR "electronic health" OR "electronic intervention" OR "digital intervention" OR "electronic tool" OR "digital tool" OR "digital technology" OR "electronic technology" OR "digital technologies" OR "digital health" OR "electronic technologies" OR "e-communication" OR "e-intervention" OR telecounseling OR telemonitoring OR telemedicine OR "video visit" OR "video consult" OR "video consultation" OR telehealth OR "remote monitoring" OR "internet communication technologies" OR "internet communication tool" OR "clinical decision support" OR teleconsultation OR teleconsultations[MeSH Terms])).

Search Query for IEEE Xplore:

Search Query 1:

("Abstract":Informatics OR "Abstract":technology OR "Abstract": "Information Technology" OR "Abstract": "Electronic Health Records" OR "Abstract": "Software" OR "Abstract":Telemedicine OR "Abstract":informatics OR "Abstract": "information technologies" OR "Abstract":groupware OR "Abstract": "group ware" OR "Abstract": "computer supported cooperative work" OR "Abstract": "computer-supported cooperative work" OR "Abstract": "shared workspace" OR "Abstract": "shared work-space" OR "Abstract": "shared work space" OR "Abstract": "information and communication technology" OR "Abstract": "information & communication technology" OR "Abstract": "information and communication technologies" OR "Abstract": "information & communication technologies" OR "Abstract": "information systems" OR "Abstract": "information system" OR "Abstract": "data system" OR "Abstract": "data systems" OR "Abstract": "informational system" OR "Abstract": "computer system" OR "Abstract": "computer systems" OR "Abstract": "web-based" OR "Abstract": "internet-based" OR "Abstract": "web 2.0" OR "Abstract": "world wide web" OR "Abstract": "world-wide web" OR "Abstract": "user-computer" OR "Abstract": "human-computer" OR "Abstract": "computer interface" OR "Abstract": "computer interfaces" OR "Abstract": "health information exchange" OR "Abstract": "health information exchanges" OR "Abstract": "interoperability" OR "Abstract": "interoperable" OR "Abstract": "electronic health record" OR "Abstract": "electronic health records" OR "Abstract": "electronic medical record" OR "Abstract": "electronic medical records" OR "Abstract": "electronic patient record" OR "Abstract": "electronic patient records" OR "Abstract": "patient portal" OR "Abstract": "patient portals" OR "Abstract": "personal health record" OR "Abstract": "personal health records" OR "Abstract": "personal-health record" OR "Abstract": "personal-health records" OR "Abstract": "tethered health record" OR "Abstract": "tethered health records" OR "Abstract": mhealth OR "Abstract": "m-health" OR "Abstract": "mobile health" OR "Abstract": "touch screen" OR "Abstract": touchscreen OR "Abstract": "smart phone" OR "Abstract": ehealth OR "Abstract": "e-health" OR "Abstract": "electronic health" OR "Abstract": "electronic intervention" OR "Abstract": "digital intervention" OR "Abstract": "electronic tool" OR "Abstract": "digital tool" OR "Abstract": "digital technology" OR "Abstract": "electronic technology" OR "Abstract": "digital technologies" OR "Abstract": "digital health" OR "Abstract": "electronic technologies" OR "Abstract": "e-communication" OR "Abstract": "e-intervention" OR "Abstract": telecounseling OR "Abstract": telemonitoring OR "Abstract": telemedicine OR "Abstract": "video visit" OR "Abstract": "video consult" OR "Abstract": "video consultation" OR "Abstract": telehealth OR "Abstract": "remote monitoring" OR "Abstract": "internet communication technologies" OR "Abstract": "internet communication tool" OR "Abstract": "clinical decision support" OR "Abstract": teleconsultation OR "Abstract": teleconsultations) AND ("Abstract":amnesia OR "Abstract":dementia OR "Abstract": "mild cognitive impairment" OR "Abstract":alzheimer OR "Abstract": "memory issue" OR "Abstract": "memory problem" OR "Abstract": "memory disorder") AND ("Abstract":treatment OR "Abstract":medication OR "Abstract":drug OR "Abstract":patient OR "Abstract":medical) AND ("Abstract":adherence OR "Abstract":compliance OR "Abstract":non-adherence OR "Abstract":persistence OR "Abstract":taking OR "Abstract":engagement).

Search Query 2:

("Publication Title":Informatics OR "Publication Title":technology OR "Publication Title": "Information Technology" OR "Publication Title": "Electronic Health Records" OR "Publication Title": "Software" OR "Publication Title":Telemedicine OR "Publication Title":informatics OR "Publication Title": "information technologies" OR "Publication Title":groupware OR "Publication Title": "group ware" OR "Publication Title": "computer supported cooperative work" OR "Publication Title": "computer-supported cooperative work" OR "Publication Title": "shared workspace" OR "Publication Title": "shared work-space" OR "Publication Title": "shared work space" OR "Publication Title": "information and communication technology" OR "Publication Title": "information & communication technology" OR "Publication Title": "information and communication technologies" OR "Publication Title": "information & communication technologies" OR "Publication Title": "information systems" OR "Publication Title": "information system" OR "Publication Title": "data system" OR "Publication Title": "data systems" OR "Publication Title": "informational system" OR "Publication Title": "computer system" OR "Publication Title": "computer systems" OR "Publication Title": "web-based" OR "Publication Title": "internet-based" OR "Publication Title": "web 2.0" OR "Publication Title": "world wide web" OR "Publication Title": "world-wide web" OR "Publication Title": "user-computer" OR "Publication Title": "human-computer" OR "Publication Title": "computer interface" OR "Publication

Title": "computer interfaces" OR "Publication Title": "health information exchange" OR "Publication Title": "health information exchanges" OR "Publication Title": "interoperability" OR "Publication Title": "interoperable" OR "Publication Title": "electronic health record" OR "Publication Title": "electronic health records" OR "Publication Title": "electronic medical record" OR "Publication Title": "electronic medical records" OR "Publication Title": "electronic patient record" OR "Publication Title": "electronic patient records" OR "Publication Title": "patient portal" OR "Publication Title": "patient portals" OR "Publication Title": "personal health record" OR "Publication Title": "personal health records" OR "Publication Title": "personal-health record" OR "Publication Title": "personal-health records" OR "Publication Title": "tethered health record" OR "Publication Title": "tethered health records" OR "Publication Title": "mhealth" OR "Publication Title": "m-health" OR "Publication Title": "mobile health" OR "Publication Title": "touch screen" OR "Publication Title": "touchscreen" OR "Publication Title": "smart phone" OR "Publication Title": "ehealth" OR "Publication Title": "e-health" OR "Publication Title": "electronic health" OR "Publication Title": "electronic intervention" OR "Publication Title": "digital intervention" OR "Publication Title": "electronic tool" OR "Publication Title": "digital tool" OR "Publication Title": "digital technology" OR "Publication Title": "electronic technology" OR "Publication Title": "digital technologies" OR "Publication Title": "digital health" OR "Publication Title": "electronic technologies" OR "Publication Title": "e-communication" OR "Publication Title": "e-intervention" OR "Publication Title": "telecounseling" OR "Publication Title": "telemonitoring" OR "Publication Title": "telemedicine" OR "Publication Title": "video visit" OR "Publication Title": "video consult" OR "Publication Title": "video consultation" OR "Publication Title": "telehealth" OR "Publication Title": "remote monitoring" OR "Publication Title": "internet communication technologies" OR "Publication Title": "internet communication tool" OR "Publication Title": "clinical decision support" OR "Publication Title": "teleconsultation" OR "Publication Title": "teleconsultations" AND ("Publication Title": "amnesia" OR "Publication Title": "dementia" OR "Publication Title": "mild cognitive impairment" OR "Publication Title": "alzheimer" OR "Publication Title": "memory issue" OR "Publication Title": "memory problem" OR "Publication Title": "memory disorder") AND ("Publication Title": "treatment" OR "Publication Title": "medication" OR "Publication Title": "drug" OR "Publication Title": "patient" OR "Publication Title": "medical") AND ("Publication Title": "adherence" OR "Publication Title": "compliance" OR "Publication Title": "non-adherence" OR "Publication Title": "persistence" OR "Publication Title": "taking" OR "Publication Title": "engagement").

Search Query for Web Of Science

Informatics OR technology OR "Information Technology" OR "Electronic Health Records" OR "Software" OR Telemedicine OR informatics OR "information technologies" OR groupware OR "group ware" OR "computer supported cooperative work" OR "computer-supported cooperative work" OR "shared workspace" OR "shared work-space" OR "shared work space" OR "information and communication technology" OR "information & communication technology" OR "information and communication technologies" OR "information & communication technologies" OR "information systems" OR "information system" OR "data system" OR "data systems" OR "informational system" OR "computer system" OR "computer systems" OR "web-based" OR "internet-based" OR "web 2.0" OR "world wide web" OR "world-wide web" OR "user-computer" OR "human-computer" OR "computer interface" OR "computer interfaces" OR "health information exchange" OR "health information exchanges" OR "interoperability" OR "interoperable" OR "electronic health record" OR "electronic health records" OR "electronic medical record" OR "electronic medical records" OR "electronic patient record" OR "electronic patient records" OR "patient portal" OR "patient portals" OR "personal health record" OR "personal health records" OR "personal-health record" OR "personal-health records" OR "tethered health record" OR "tethered health records" OR mhealth OR "m-health" OR "mobile health" OR "touch screen" OR touchscreen OR "smart phone" OR ehealth OR "e-health" OR "electronic health" OR "electronic intervention" OR "digital intervention" OR "electronic tool" OR "digital tool" OR "digital technology" OR "electronic technology" OR "digital technologies" OR "digital health" OR "electronic technologies" OR "e-communication" OR "e-intervention" OR telecounseling OR telemonitoring OR telemedicine OR "video visit" OR "video consult" OR "video consultation" OR telehealth OR "remote monitoring" OR "internet communication technologies" OR "internet communication tool" OR "clinical decision support" OR teleconsultation OR teleconsultations (Topic) and adherence OR compliance OR non-adherence OR persistence OR taking OR engagement (Topic) and treatment OR medication OR drug OR patient OR medical (Topic) and amnesia OR dementia OR "mild cognitive impairment" OR alzheimer OR "memory issue" OR "memory problem" OR "memory disorder."

Search Query for ProQuest

summary(Informatics OR technology OR "Information Technology" OR "Electronic Health Records" OR "Software" OR Telemedicine OR informatics OR "information technologies" OR groupware OR "group ware" OR "computer supported cooperative work" OR "computer-supported cooperative work" OR "shared workspace" OR "shared work-space" OR "shared work space" OR "information and communication technology" OR "information & communication technology" OR "information and communication technologies" OR "information & communication technologies" OR "information systems" OR "information system" OR "data system" OR "data systems" OR "informational system" OR "computer system" OR "computer systems" OR "web-based" OR "internet-based" OR "web 2.0" OR "world wide web" OR "world-wide web" OR "user-computer" OR "human-computer" OR "computer interface" OR "computer interfaces" OR "health information exchange" OR "health

information exchanges" OR "interoperability" OR "interoperable" OR "electronic health record" OR "electronic health records" OR "electronic medical record" OR "electronic medical records" OR "electronic patient record" OR "electronic patient records" OR "patient portal" OR "patient portals" OR "personal health record" OR "personal health records" OR "personal-health record" OR "personal-health records" OR "tethered health record" OR "tethered health records" OR mhealth OR "m-health" OR "mobile health" OR "touch screen" OR touchscreen OR "smart phone" OR ehealth OR "e-health" OR "electronic health" OR "electronic intervention" OR "digital intervention" OR "electronic tool" OR "digital tool" OR "digital technology" OR "electronic technology" OR "digital technologies" OR "digital health" OR "electronic technologies" OR "e-communication" OR "e-intervention" OR telecounseling OR telemonitoring OR telemedicine OR "video visit" OR "video consult" OR "video consultation" OR telehealth OR "remote monitoring" OR "internet communication technologies" OR "internet communication tool" OR "clinical decision support" OR teleconsultation OR teleconsultations) AND summary(amnesia OR dementia OR "mild cognitive impairment" OR alzheimer OR "memory issue" OR "memory problem" OR "memory disorder") AND summary (treatment OR medication OR drug OR patient OR medical) AND summary(adherence OR compliance OR non-adherence OR persistence OR taking OR engagement)."



THIEME

Supplementary Appendix C Quality assessment results using the MMAT tool

Title (Ref. and year)	Screening questions		Quantitative randomized					Quantitative non-RC					Final score	Rating ^a
	Are there clear research questions?	Do the collected data allow to address the research questions?	Is randomization appropriately performed?	Are the groups comparable at baseline?	Are there complete outcome data?	Are outcome assessors blinded to the intervention provided?	Did the participants adhere to the assigned intervention?	Are the participants representative of the target population?	Are measurements appropriate regarding both the outcome and intervention (or exposure)?	Are there complete outcome data?	Are the confounders accounted for in the design and analysis?	During the study period, is the intervention administered (or exposure occurred) as intended?		
42 2006	Y	Y	Y	Y	Y	Y	U						80%	B
43 2015	Y	Y	Y	Y	Y	Y	Y						100%	A
34 2016	Y	Y						N	Y	Y	Y	Y	80%	B
46 2016	Y	Y	Y	Y	Y	Y	Y						100%	A
35 2017	Y	Y	Y	Y	Y	Y	Y						100%	A
33 2019	Y	Y	Y	Y	Y	Y	N						80%	B
47 2019	Y	Y	Y	Y	Y	Y	N						80%	B
41 2019	Y	Y						N	Y	Y	Y	Y	80%	B
40 2020	Y	Y	Y	Y	Y	Y	Y						100%	A
36 2020	Y	Y	Y	Y	Y	Y	N						80%	B
39 2020	Y	Y	Y	Y	Y	Y	U						80%	B
34 2016	Y	Y	Y	Y	Y	Y	Y						100%	A
46 2016	Y	Y	Y	Y	Y	Y	Y						100%	A
37 2021	Y	Y	Y	Y	Y	Y	N						80%	B
38 2021	Y	Y						N	Y	Y	Y	Y	80%	B
44 2022	Y	Y	Y	Y	Y	Y	Y						100%	A
45 2022	Y	Y	Y	Y	Y	Y	Y						100%	A

Abbreviations: MMAT, Mixed Methods Appraisal Tool; N, no; U, unclear; Y, yes.

^aA: low bias, B: moderate bias, C: high bias.

Author Query Form (ACI/202308R0191)

Special Instructions: Author please write responses to queries directly on proofs and then return back.

Q1: AU: Please confirm that given names (red), middle names (black) and surnames (green) have been identified correctly. Author names in bibliographic citations and online will appear as: Elkefi S, Blecker S, Bitan Y. Health Information Technology Supporting Adherence Memory Disorder Patients: A Systematic Literature Review. Please confirm if this is correct.

Q2: AU: Please provide division/department name in affiliations 1–3.

Q3: AU: Please check city and state names for accuracy, and provide zip code in the address for correspondence.

Q4: AU: The webpage associated with the URL provided is not accessible. Please check and provide the correct URL.

Q5: AU: References 10 & 13, 9 & 33 were same. Ref. 9 and 10 retained and 13 and 33 deleted from reference list and renumbered accordingly.

Q6: AU: Ref. 1 has been updated as per PubMed database. Please check.

Q7: AU: Please check the inserted URL in Ref. 5 for accuracy.

Q8: AU: Ref. 11 has been updated. Please check and provide page range.

Q9: AU: Please update Ref. 24, if already published.

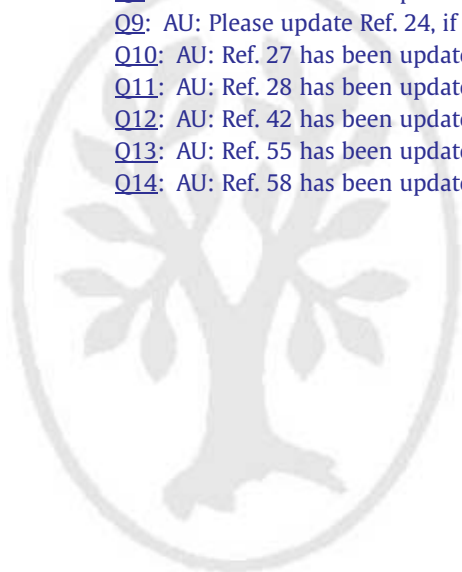
Q10: AU: Ref. 27 has been updated. Please check.

Q11: AU: Ref. 28 has been updated as per PubMed database. Please check.

Q12: AU: Ref. 42 has been updated as per PubMed database. Please check.

Q13: AU: Ref. 55 has been updated as per PubMed database. Please check.

Q14: AU: Ref. 58 has been updated. Please check.



THIEME